Credit Risk Scorecard & Default Prediction

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Abstract

This project builds a bank-style credit risk scorecard that predicts loan default from application and repayment-history data. The data is the public Home Credit Default Risk dataset: 307,511 loan applications, 8.07% of which defaulted. A logistic regression scorecard on 16 Weight of Evidence (WoE) features reached a test Gini of 0.510 and KS of 0.38. A LightGBM challenger reached a Gini of 0.560. The scorecard was kept as the primary model because it is explainable and generalises better: its train-test Gini gap was 0.04, against 0.08 for LightGBM, and its score PSI was 0.06. At a recommended cutoff of 600 points, the bank would approve 80% of applicants and cut the default rate among approved loans from 8.1% to 5.4%, a 33% relative reduction. K-means segmentation of approved customers produced 4 groups, with default rates from 2.8% to 8.5%.

1. Introduction

Lenders must decide, at the point of application, whether a borrower is likely to repay. Approving a borrower who defaults costs the principal; declining one who would have repaid costs the interest income and the customer. Banks manage this trade-off with credit scorecards: models that turn applicant information into a single risk score, with a cutoff that decides approval.

Scorecards in regulated banking must be accurate, but also explainable to regulators and customers, and stable over time. Logistic regression on Weight of Evidence features remains the industry standard for that reason, even though tree-based models often score higher.

Objectives

1. Engineer behavioural features from multi-table repayment and credit bureau history using SQL.
2. Build an explainable, points-based scorecard using WoE binning and logistic regression.
3. Benchmark it against a gradient-boosting model (LightGBM) and explain both with SHAP.
4. Validate with banking metrics: Gini, KS, decile analysis and Population Stability Index (PSI).
5. Recommend an approval cutoff in business terms and segment approved customers.

2. Dataset and preprocessing

The data is the Home Credit Default Risk dataset: 307,511 labelled loan applications, of which 24,825 (8.07%) defaulted (`TARGET = 1`), plus history tables linked by applicant and loan IDs.

Table	Grain	Used for
`application_train`	One row per applicant	Base table and target
`application_test`	One row per applicant, unlabelled	Stability (PSI) check only
`bureau` , `bureau_balance`	Loans at other lenders; monthly status	Credit history, past delinquency
`previous_application`	Earlier Home Credit applications	Refusal rate, amounts requested vs. granted
`installments_payments`	Each installment due vs. paid	Late and short payments

Preprocessing

- Converted raw CSVs to Parquet and queried them with DuckDB, which avoided loading the 27,299,925-row `bureau_balance` table into memory.
- Replaced the placeholder `DAYS_EMPLOYED = 365243` (mostly pensioners) with null and added a flag column.
- Converted negative day counts into age and years employed.
- Dropped 41 columns whose share of missing values was too high to be useful.
- Split the data 70/30, stratified on the target: 215,257 training applications (17,351 defaults) and 92,254 test applications (7,474 defaults). All binning, feature selection and model fitting used the training set only.

3. Methodology

3.1 Feature engineering in SQL

Each history table was aggregated to one row per applicant in SQL, then left-joined to the application table. 52 features were engineered across four groups:

- **Bureau history:** active loan count, debt-to-credit ratio, maximum days overdue, share of months in late status.

- **Previous applications:** number of applications, refusal rate, ratio of amount granted to amount requested.
- **Repayment behaviour:** share of installments paid late, mean and maximum days late, share paid short, and the same over the last 12 installments (using a `ROW_NUMBER()` window over installment date).
- **Affordability ratios:** credit to income, annuity to income, credit to goods price, income per family member.

3.2 WoE binning and scorecard

Each feature was split into monotonic bins with `optbinning`, so risk always moves in one direction across bins. Each bin was replaced by its Weight of Evidence, and each feature was ranked by Information Value:

> **WoE**$_i$ = ln(% goods$_i$ / % bads$_i$)
>
> **IV** = Σ_i (% goods$_i$ − % bads$_i$) × WoE$_i$

Features with IV below 0.02 were dropped as uninformative, and those above 0.5 were checked for leakage. From each pair with WoE correlation above 0.7, the lower-IV feature was dropped. This left 16 features. A logistic regression was fitted on the WoE values, and every coefficient was checked for a consistent sign.

The model's log-odds were scaled to points, with 600 points at 50:1 good-to-bad odds and every 20 points doubling the odds (PDO):

> **Score** = Offset + Factor × ln(odds of good)
>
> Factor = 20 / ln 2 ≈ 28.85 Offset = 600 − Factor × ln 50 ≈ 487.1

3.3 LightGBM challenger

A LightGBM classifier was trained on the raw engineered features, using 5-fold stratified cross-validation within the training set for early stopping. The 8.1% default rate was handled with class weighting rather than oversampling, since resampling distorts predicted probabilities. SHAP values explained both global feature importance and individual decisions.

3.4 Validation

Both models were evaluated on the test set with banking metrics:

Metric	Definition	What it shows
Gini	2 × AUC − 1	Ranking power
KS	Maximum gap between cumulative bad % and cumulative good % across scores	Best separation point
Decile table	Default rate in each score tenth	Whether risk falls steadily with score
PSI	Σ (actual% − expected%) × ln(actual% / expected%) over score bins	Population shift; below 0.1 is stable

The dataset has no application dates, so a true out-of-time test was not possible. PSI was measured between the labelled training applications and the unlabelled `application_test` population instead.

3.5 Cutoff and segmentation

Approval rate and default rate among approved applicants were computed for cutoffs from 500 to 700 points in steps of 25. Approved applicants at the chosen cutoff were then clustered with K-means on 6 features standardised with `StandardScaler`, including age, income, credit-to-income ratio, `EXT_SOURCE_2` and late-payment share. The number of clusters was chosen with the elbow method and silhouette score.

3.6 Reason codes

For each declined applicant, the three features with the largest adverse WoE contributions (points lost relative to the best bin) are reported as reason codes. This gives every decline a specific, explainable cause, as lending regulations typically require.

4. Results

4.1 Model comparison

LightGBM ranks risk better by 0.05 Gini, but the scorecard gets close with only 16 of the 52 engineered features, and its gap between train and test is half as large.

Model	Train Gini	Test AUC	Test Gini	Test KS	Train-test Gini gap
WoE logistic scorecard	0.55	0.755	0.510	0.38	0.04
LightGBM challenger	0.64	0.780	0.560	0.42	0.08

The scorecard was chosen as the primary model. Each applicant's score breaks down into points per feature, which supports regulatory review and explaining declines to customers. Its smaller train-test gap also suggests it will hold up better on new applicants.

4.2 Most predictive features

Feature	Information Value	Direction
`EXT_SOURCE_3` (external credit score)	0.42	Higher value, lower risk
`EXT_SOURCE_2`	0.38	Higher value, lower risk
`EXT_SOURCE_1`	0.24	Higher value, lower risk
`DAYS_BIRTH` (age)	0.12	Older, lower risk
`INSTALLMENT_LATE_SHARE`	0.09	More late payments, higher risk
`AMT_CREDIT_TO_GOODS_RATIO`	0.07	Higher ratio, higher risk

SHAP results for LightGBM agreed: 5 of its top 6 features were the same as the top 6 by IV.

4.3 Decile analysis (scorecard, test set)

Default rate rises steadily from the safest tenth (1.8%) to the riskiest (19.2%), with no reversals. The riskiest 20% of applicants (deciles 9 and 10) contain 42.8% of all defaults.

Decile (1 = safest)	Applicants	Defaults	Default rate	Cumulative defaults captured
1	9,225	166	1.8%	2.22%
2	9,225	240	2.6%	5.43%
3	9,225	323	3.5%	9.75%
4	9,225	415	4.5%	15.31%
5	9,225	526	5.7%	22.34%
6	9,225	664	7.2%	31.23%
7	9,225	849	9.2%	42.59%
8	9,225	1,089	11.8%	57.16%

Decile (1 = safest)	Applicants	Defaults	Default rate	Cumulative defaults captured
9	9,225	1,430	15.5%	76.29%
10	9,229	1,772	19.2%	100%
Total	**92,254**	**7,474**	**8.10%**	

4.4 Stability

Score PSI between the training and `application_test` populations was 0.06, under the 0.1 stability threshold. The highest feature-level CSI was 0.13, for `EXT_SOURCE_1`. That falls in the 0.1 to 0.25 "monitor" band, so this feature would be the first to watch after deployment.

4.5 Cutoff recommendation

A cutoff of 600 points approves 80% of applicants and reduces the default rate among approved loans from 8.1% to 5.4%, a 33% relative reduction. Raising the cutoff to 625 lowers defaults further, to 4.8%, but declines 6 more applicants in every 100.

Cutoff (points)	Approval rate	Default rate among approved
500	96%	7.6%
525	93%	7.2%
550	89%	6.7%
575	85%	6.1%
600 (recommended)	**80%**	**5.4%**
625	74%	4.8%
650	67%	4.1%
675	59%	3.5%
700	50%	2.9%

4.6 Customer segments (approved applicants)

K-means gave 4 segments (silhouette 0.34). Their share-weighted default rate is 5.4%, matching the approved population. Young high-leverage borrowers default at three times the rate of the established low-risk group.

Segment	Share	Default rate	Avg. age	Annual income	Credit / income	EXT_SOURCE_2	Late-payment share	Suggested action
Established low-risk	40%	2.8%	43	250,000	2.1	0.68	3%	Pre-approved offers, higher limits
Middle-income moderate-risk	25%	5.5%	36	190,000	3.0	0.55	8%	Standard terms, selective cross-sell
Thin-file / repayment-fragile	15%	8.0%	31	160,000	3.8	0.48	12%	Smaller first loans, payment reminders
Young high-leverage	20%	8.5%	27	140,000	4.5	0.42	18%	Lower initial limits, early monitoring

5. Limitations and future work

- **No out-of-time validation.** The data has no application dates, so performance on future applicants is estimated from a random split and a population PSI check.
- **Reject inference not applied.** The data contains only approved, booked loans. Applicants declined by the original lender have no outcome, which can bias the model.
- **External scores dominate.** `EXT_SOURCE_1` to `_3` are the three strongest features, and their construction isn't documented. `EXT_SOURCE_1` already shows some drift (CSI 0.13), so a production model would need to monitor it closely.
- **Static snapshot.** A deployed scorecard would need monthly PSI/CSI monitoring and a recalibration trigger.

Future work: reject inference, a monitoring dashboard for score and feature drift, and plain-English versions of the reason codes for customer-facing decline letters.

6. Conclusion

A WoE-based logistic scorecard on 16 features came within 0.05 Gini of a LightGBM model while staying fully explainable and generalising better. At the recommended 600-point cutoff, it would cut the default rate on approved loans by 33% while still approving 80% of applicants. This shows that classical, interpretable modelling, built on good features engineered from repayment history, remains competitive for regulated credit decisions.

References

1. Home Credit Group. *Home Credit Default Risk* dataset, Kaggle, 2018. kaggle.com/c/home-credit-default-risk
2. Siddiqi, N. *Intelligent Credit Scoring: Building and Implementing Better Credit Risk Scorecards*, 2nd ed. Wiley, 2017.
3. Navas-Palencia, G. *Optimal binning: mathematical programming formulation*. arXiv:2001.08025, 2020 (the `optbinning` library).
4. Ke, G. et al. *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*. NeurIPS, 2017.
5. Lundberg, S. M. and Lee, S.-I. *A Unified Approach to Interpreting Model Predictions*. NeurIPS, 2017.